

# **Challenges and Success Factors of Informal AI Upskilling: A Quantitative Study Among Non-IT Employees**

## **Executive Summary**

### **Motivation**

Artificial intelligence tools like Copilot and ChatGPT have become part of Daily work for many employees. Organizations are adopting these tools at a growing rate and expect them to be utilized by their employees. Formal training on how to better use them however, does not keep up with the speed of this adoption. As a result, the majority is left to build their AI skills informally while doing their jobs.

Most of the existing research on AI related skill development focuses on IT professionals or on employees who already received structured training. However, employees outside of IT departments make up the majority of employees in many industries which the focus is not technology. The knowledge on what influences their informal AI learning is limited.

Understanding this problem and suggesting possible solutions has clear value for organizations. If they know which factors support informal AI upskilling and which factors block it, they can build better conditions and eliminate the barriers on it. As a result, employees would be able to learn these promising tools and benefit from them more.

### **What We Already Know**

Prior research shows that employees learn most of their job skills informally, not through formal training programs. This holds true across many industries and has been confirmed for AI adoption specifically. Studies on workplace learning have identified motivation, learning opportunities, social support and organizational environment as solid drivers of informal learning.

Several established theories help explain how people adopt and use new technologies. The Unified Theory of Acceptance and Use of Technology (UTAUT) explains technology adoption through perceived usefulness, perceived ease of use, social influence, and facilitating conditions. The Job Demands-Resources model, or JD-R, explains how job demands drain energy while job resources support engagement and growth. Social learning theory explains how people learn by observing and imitating others, especially peers they see as similar to themselves. More recent research on AI identity shows that people can develop a personal, professional connection to AI that goes beyond simple usefulness or ease of use.

### **What We Don't Know**

We still do not know which factors matter most for informal AI upskilling among non-IT employees specifically. Existing studies mostly focus on IT professionals or on employees who already received formal training.

We also do not know how these different theories relate to each other in the context of AI upskilling. UTAUT, JD-R, social learning theory, and AI identity research have been studied separately. No study has been found that combines all four into a single model to explain

informal AI upskilling for non-IT employees even though they are closely related when looking from a technology learning perspective.

### **Research Question**

This thesis has three research questions. The first asks how perceived usefulness and perceived ease of use from UTAUT model relate to informal AI upskilling. The second asks how technology overload as a job demand and managerial support as a job resource relate to informal AI upskilling. The third asks what role peer influence and AI identity play in informal AI upskilling.

In its essence these three questions can be summarized in one with which individual-level and organizational-level factors actually predict the intensity of informal AI upskilling among non-IT employees?

### **Research Model and Theoretical Background**

The thesis combines four theoretical perspectives into a single model. These perspectives are UTAUT, the JD-R model, social learning theory and AI identity research.

The conceptual model built in the thesis separates six predictors into two levels. The individual level includes perceived usefulness, perceived ease of use, peer influence, and AI identity. The organizational level includes technology overload and managerial support. Additionally, control variables were also included for better understanding. These include age, gender, education, work experience, prior AI experience, and prior formal AI training.

Six hypotheses connect these predictors to the dependent variable of informal AI upskilling intensity:

- H1: Perceived usefulness is positively related to informal AI upskilling intensity.
- H2: Perceived ease of use is positively related to informal AI upskilling intensity.
- H3: Peer influence is positively related to informal AI upskilling intensity.
- H4: AI identity is positively related to informal AI upskilling intensity.
- H5: Technology overload is negatively related to informal AI upskilling intensity.
- H6: Managerial support is positively related to informal AI upskilling intensity.

The dependent variable is measured with an AI adapted version of the Informal Workplace Learning scale developed by Decius et al. (2019). This adapted scale captures how often employees experiment, learn from colleagues, reflect on their own experiences and look for new ways to apply AI at their daily work.

### **Empirical Approach to Test the Hypotheses**

To test these hypotheses, the thesis uses a quantitative survey design. The survey was distributed to non-IT employees at a multinational medical technology company with about 7,000 employees worldwide.

The survey was sent to 271 employees across three groups after necessary permissions were taken. These were global corporate finance staff, Germany-based internal staff, and Germany-based external staff. In total, 111 employees completed the survey, giving a response rate of 41%.

All constructs were measured using established, validated scales, adapted for the AI context. Responses were collected on a five-point Likert scale, from strongly disagree to strongly agree for independent variables and from never to always for the dependent variable.

The collected data was analyzed in JASP using two regression models. The first model tested all six predictors together, without any control variables. The second model was a hierarchical regression. It added control variables first, then individual-level predictors, then organizational-level predictors. This step by step approach was used to see how much each block of variables adds to the explanation of informal AI upskilling.

## Results

The sample was balanced with 55% men and 45% women. Most respondents were from the global finance group as it had the highest amount of members within the groups survey was shared with. Second and third most responses came from sales and marketing departments. All measurement scales passed reliability tests with Cronbach's alpha values ranging between 0.78 and 0.89.

The six predictors together explained a large share of the variance in informal AI upskilling. In the first regression without control variables, the model had a  $R^2$  value of 0.64. In the second regression with control variables  $R^2$  increased to 0.75.

Two hypotheses were supported in the full regression model. Peer influence had a significant positive relationship with informal AI upskilling with  $\beta = 0.18$  and  $p = .014$ . AI identity had the strongest relationship of any predictor tested in the entire study with  $\beta = 0.35$  and  $p < .001$ .

The remaining four hypotheses were not supported once control variables were included. Perceived usefulness lost almost all of its significance with  $\beta = -0.01$  and  $p = .938$ , even though it was significant on its own in the model without controls. Perceived ease of use was close to significance but did not reach with  $\beta = 0.15$  and  $p = .095$ . Technology overload also came close but did not reach significance with  $\beta = -0.13$  and  $p = .053$ . Managerial support showed no significant relationship in either model with  $\beta = 0.06$  and  $p = .362$ .

Among the control variables, prior AI experience was the strongest predictor of informal AI upskilling with  $\beta = 0.36$  and  $p < .001$ . Gender was also significant with  $\beta = -0.13$  and  $p = .019$ , with men reporting somewhat higher informal AI upskilling than women in this sample. However, the gender result should be read with caution, given the sample size of the study.

## Discussion

The results suggest that individual, social, and identity-related factors matter more for informal AI upskilling than technology perceptions or organizational conditions. Employees who see AI as part of their professional identity are more likely to keep developing their AI skills on their own. The same is true for employees who learn from trusted colleagues and see their peers making an effort in learning how to work with AI tools.

Perceived usefulness and perceived ease of use still matter in a general sense. Their effect, however, crosses heavily with prior AI experience. Employees who have used AI for longer also are more likely to see it as more useful and easier to use. Once prior experience is

included in the model, these two constructs from the Technology Acceptance Model add little unique explanation on their own.

Technology overload showed a relationship in the expected direction. It did not, however, reach significance in the full model with a slight difference. This suggests the relationship may exist, but at a bigger sample size than this sample could detect it better.

Managerial support was not significant in either of the models. One possible explanation is that informal learning is, by definition, something employees choose to do on their own. Managerial encouragement may matter more for formal training programs than for informal, self-directed learning. Adding technology overload and managerial support together only explained an additional 1% of the variance after the individual-level predictors, which shows a limited effect for the organizational level in this specific type of learning behavior.

### **Implications for Research and Practice**

An important implication for research contributed by this thesis is that UTAUT, JD-R, social learning theory, and AI identity research can be combined into one useful model. It also shows that AI identity deserves more attention in future studies on technology adoption, since it is a newly developing concept. Neither UTAUT nor JD-R alone would have captured the importance of AI identity and peer influence found in this thesis.

Practical implications for organizations that want to support informal AI upskilling should focus less on convincing employees that AI is useful. Instead of this, organizations should build environments where employees can see AI as part of their professional identity, and where they can learn from one another.

Additionally, general AI training sessions did not show meaningful relationship with informal upskilling in this sample. Role-specific trainings and visible peer learning, such as identifying colleagues who are already experienced AI users as informal points of contact, may work better for a more developed employee base with AI skills.

### **Conclusion**

This thesis aimed to identify which individual and organizational factors predict informal AI upskilling among non-IT employees. The results show that AI identity and peer influence are the two clearest success factors. Technology perceptions, technology overload, and managerial support play a smaller or non-significant role, once other factors are considered.

Organizations that want employees to keep developing their AI skills on their own should invest in building a real professional connection to AI for their employees. They should also support peer learning by encouraging knowledge sharing on best practices within using AI tools.

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